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EDITORIAL

The biological roots of political division: mapping the neural architecture of ideology and social influence

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For a long time, political thought was shaped by the image of the "rational voter," someone who calmly evaluates policies and candidates before making a choice. Research in political psychology has steadily challenged that picture. Even so, the biological and cognitive processes that shape political identities, susceptibility to misinformation, and emotional reactions to leaders are only now coming into sharper focus. This Collection brings together recent work that goes beyond self-report and draws on neuroimaging, computational modelling, and behavioral interventions to examine the architecture of the political mind. Across these studies, political behavior appears closely linked to basic sensory, affective, and cognitive processes. Here, we synthesize those contributions and argue that future progress in political neuroscience will depend on stronger predictive computational models. Rather than treating ideology as either fully rational or simply irrational, this perspective suggests that political beliefs often emerge from ordinary neural systems trying to cope with complexity, uncertainty, and threat.

We live in a period of striking political instability. Polarization is deepening across many democracies, trust in institutions continues to weaken, and misinformation spreads easily through fragmented media environments. To understand these broad social changes, researchers are increasingly examining the underlying mechanisms of perception, attention, learning, and decision making. This Collection brings neuroscience, psychology, and political science into direct conversation. Its central message is simple: political reality is shaped not only by institutions and campaigns but also by the rapid, often implicit operations of the human brain. Looking more closely at those processes may help us understand both the origins and the consequences of political division in a new way.

The neural signatures of "us versus them"

One of the fundamental questions in political psychology is how people perceive political allies and opponents. Does polarization reflect emotional disengagement from the other side, or something more active? Homan et al.[1] offer an important clue through their study of the mu rhythm, a neural oscillation involved in processing the actions and emotions of other people. Their findings do not support the idea that people simply fail to empathize with the out party. Instead, participants showed stronger neural engagement when viewing the emotional expressions, especially anger, of politicians they opposed. In other words, the polarized brain may not ignore opponents at all. It may monitor them more intensely, as if they are especially salient or potentially threatening.

This kind of neural differentiation also appears in the way political messages are absorbed. In a study of extremist recruitment, Cohen et al.[2] used fMRI to distinguish the neural pathways engaged by different forms of propaganda. "Heroic" narratives, which promised personal glory, were associated with activity in the ventral striatum, a core part of the brain's reward system. "Social" narratives, which emphasized obligation to the group, engaged the dorsomedial prefrontal cortex, a region linked to mentalizing. The contrast is revealing. Political

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persuasion is not a single process. Different messages work through different neural routes, depending on what the audience is seeking or valuing.

Uncertainty, stress, and the switch to rigidity

As polarization intensifies, conspiracy beliefs and dogmatic forms of thinking often become more attractive. Two papers in this Collection help explain why, and they do so by focusing on how the brain allocates limited cognitive resources under uncertainty and stress.

Leclercq et al.[3] present a computational account of how political uncertainty can increase susceptibility to conspiratorial beliefs. Using a Circular Inference model fitted to behavioral data collected during stressful political periods, including the U.S. elections and Brexit, they show that uncertainty first pushes people toward overweighting incoming sensory information. This resembles an exploratory mode. Later, as stress subsides, the balance shifts toward an excessive reliance on prior beliefs. That shift encourages rigidity and can stabilize conspiracy ideation.

Németh et al.[4] develop this issue further in their Threat-based Neural Switch Theory. Their framework proposes that sociopolitical threat and resource scarcity bias the brain away from goal-directed processing, typically supported by prefrontal systems, and toward more habitual forms of processing associated with subcortical regions such as the striatum. Under stress, people may therefore become more receptive to simple, dogmatic messages and less able or willing to engage with complexity. Taken together, the papers by[3] and[4] suggest that conspiracy beliefs should not be dismissed as mere irrationality. They may instead reflect basic neural systems responding in maladaptive ways to uncertainty and threat.

Implicit biases and the limits of neural prediction

Neurobiological measures can reveal mechanisms that are difficult to detect through self-report alone, but they can also expose a gap between implicit processing and overt political behavior. Mahieux et al.[5] used the N400 event-related potential, a neural marker of semantic incongruity, to examine voting in the 2022 Texas gubernatorial election. The N400 captured individual policy preferences quite well. For instance, a Democrat's brain was more likely to register a kind of semantic mismatch when processing a Republican policy statement. Yet these neural signals were less effective than explicit measures of authoritarianism and partisan identity in predicting the final vote. This finding is important. It suggests that personal preferences may be present at an implicit neural level, while the final act of voting is often dominated by stronger group-based identities.

Political bias also reaches into very subtle nonverbal cues. Barragar et al[6] examined lateral posing bias and found an intriguing mismatch between politician behavior and voter preference. Politicians often present their left cheek, which has been associated with greater emotional expressivity. Voters, however, preferred candidates shown from the right cheek. The result highlights how basic perceptual asymmetries can shape political judgments outside conscious awareness. Even seemingly minor visual details may influence impressions of competence, warmth, or trustworthiness.

Intervening in the autocratic information ecosystem

Orosz et al.[7] address a related problem in the context of Hungary's informational autocracy, where loyalty to political identity can shape judgments of truth. They found that pro-government voters were less accurate than opposition voters in distinguishing fake news from real news. At the same time, the study points toward possible remedies. A wise intervention built around family-based prosocial motivation, encouraging younger people to help older relatives navigate the news environment, produced lasting gains in discernment of misinformation. This is encouraging. The cognitive roots of misinformation vulnerability may run deep, but they are not fixed. Socially meaningful interventions can sometimes soften partisan defenses and reopen space for critical reflection.

Future directions: from explanation to prediction

The studies in this Collection mark real progress in our understanding of the neurocognitive foundations of political behavior. Still, the field faces an important limitation. Much of political neuroscience remains post hoc. Researchers are often able to explain why an event happened or why a group responded in a particular way only after the fact. It is far rarer to use neural or cognitive measures to anticipate larger political shifts before they unfold. That is where the field now needs to go. Political neuroscience should move from explanation toward prediction.

Doing so will require experimental paradigms that produce data suitable for strong computational modelling. Future work should aim to develop generative models that link neural and physiological signals to subsequent changes in political attitudes and behavior at the population level. Measures such as reward-related activity in the ventral striatum or rigid prior weighting in Bayesian inference may become especially informative in this respect. As Leor Zmigrod argues in her recent book, The Ideological Brain[8], understanding cognitive flexibility and rigidity at a biological level is essential to explaining why some people become drawn toward authoritarianism while others resist it.

One can imagine a future in which neuroforecasting does more than help us interpret radicalization after it has already occurred. It may allow us to identify vulnerability earlier. If specific neural responses, such as N400 reactions to emerging policy language or amygdala reactivity to political imagery, can forecast broader shifts in public sentiment, then political neuroscience could become a genuinely predictive science. Combined with longitudinal neuroimaging and computational modelling, such an approach might help us anticipate the spread of polarizing narratives and perhaps reduce some of their most damaging effects. Any move in this direction, however, will require careful ethical safeguards so that these tools are not misused.

Conclusion

This Collection shows that political division is not only a conflict of ideas. It is also shaped by differences in the way brains perceive, evaluate, and respond to the social world. From judgments based on a politician's face to shifts in processing under threat, biological systems help shape ideological life in ways that are often invisible to us. Understanding these mechanisms will not solve polarization on its own. But it can deepen our grasp of the human mind and perhaps support better interventions for fostering more resilient, reflective, and evidence-based societies.

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Declarations

Competing Interests

The authors declare no competing interests.

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